



#14064 - Improving a Collaborative Model for Water Banking in Cache Valley, UT.

Protocol Information

Review Type	Status	Submission Number	Approval Date
Exempt	Exempt	9	Apr 06, 2026
Continuing Review Date	Expiration Date	Initial Approval Date	Initial Review Type
--	Jan 15, 2029	Mar 11, 2024	Exempt

Approval Comment

Your amendment has been approved! Please note that your official approval letter and approved consent document(s) are located in the Admin Notes and Files tab, which is available in the right-hand menu of your protocol. The only approved informed consent documents are the ones uploaded there, not the ones uploaded onto the main page of your protocol. If you ever need to view previous versions of your protocol, please refer to the "Version" drop down menu at the top of your protocol application form. The IRB wishes you luck with your research!

Feedback

Protocol Amendment Form

Amendment

Amendment Filing Instructions:

1. Identify **each area** of the protocol that will be impacted by the change you wish to make.
2. Work through your entire protocol, **editing** each field relevant to the change you are requesting. There is no need to use formatting or headers to identify your changes; Kuali makes your changes evident to reviewers.
3. Submit your amendment using the **Submit** button on the control panel on the right side of your screen. Only the Principal Investigator can submit.

Amendments can only be filed for changes you have not yet implemented. Amendments are processed in one of two ways: minor modifications are reviewed via the Expedited review process (10 or fewer business days, typically), and modifications that are more than minor must be reviewed by the Convened IRB (up to 30 days). More than minor modifications are not permitted for Exempt protocols; a new Exempt protocol must be filed if the changes are more than minor.

If any member of the research team has expired CITI training, an amendment cannot be approved until their training has been refreshed. Please be sure all research staff have up-to-date training before submitting your amendment.

Provide an overview of your amendment, including why you are requesting the change. This field is not the amendment to your protocol; it is simply an overview for reviewers. Please work through your entire protocol to ensure it reflects the changes you describe here.

Hi!

We would like to update the expiration date of this protocol and the consent documentation to Dec 31, 2028. This reflects a more reasonable time to conduct a next round of modeling sessions at the scale of the Bear River Basin. Currently the letter of information (post-it board; file name is "14064 Rosenberg LOI Posterboard AmV14 thru Mar26.pdf") has the IRB Consent Documentation expires at the top as March 1, 2026. Can we please get that updated? There is no change in the content of the post-it board.

Does the proposed change require any edits to your informed consent documentation, or the creation of a new document? We recommend briefly reviewing your approved informed consent documents, in Admin Notes & Files on the right side of your screen, to be certain.

No

Are you adding a Non-USU Investigator to this protocol?

No

Have you begun recruiting participants?

Yes, participants have been enrolled and study procedures are underway

Please review the definition of a [Conflict of Interest](#), along with the Assurances you agreed to in the Conflicts & Assurances section of this protocol.

I, the Principal Investigator, have reviewed the definition of a conflict of interest with my research team, including any new research staff I may be adding, and there are no new conflicts to disclose.

Basic Information

Principal Investigator
Rosenberg, David E

Department

Civil & Environmental Engineering

Protocol Title

Improving a Collaborative Model for Water Banking in Cache Valley, UT.

Please select your anticipated start date for this research, taking care to allow ample time for IRB review.

July 3, 2023

Is this research externally funded?

No

Will this project be used, in part or whole, for a thesis or dissertation project?

Yes

Please indicate what type of review or action you are requesting. When you make a selection, more information about the type of review will display.

Exempt Review

As you work through this protocol submission, Kuali will autosave your information every 10-20 seconds. **It is very important that you only complete this protocol in one browser/browser tab at a time.** Kuali will autosave the *last active version*, so **if your last active version open is a less complete protocol, you will lose your work.** More user documentation regarding Kuali Protocols is available on [Kuali's website](#), as well as the [USU IRB's](#).

Study Personnel

USU Personnel

Please enter all of your USU study personnel to the list below by clicking "Add Info" or "Add a Line." You may double check that they have completed CITI training [at this link](#); unless this is a Non-Human Subjects Research Determination, do not submit your protocol until all staff are CITI trained.

You only need to add personnel who will serve as an investigator on this project.

An **investigator** is an individual who interacts with or intervenes with living people for research purposes; obtains, studies, interprets, or analyzes identifiable private information; obtains informed consent; or interacts with the IRB relating to this study. Individuals assisting with the analysis of de-identified data, for example, would not be investigators under this definition.

Investigator Name

Akbar, Hadia

Role

USU Student Researcher

System Role

Admin

Permissions

Full Access

USU Personnel Email

hadia.akbar@usu.edu

Department

Utah Water Research Laboratory

CITI Training Expiration Date

1/19/2026

You only need to add personnel who will serve as an investigator on this project.

An **investigator** is an individual who interacts with or intervenes with living people for research purposes; obtains, studies, interprets, or analyzes identifiable private information; obtains informed consent; or interacts with the IRB relating to this study. Individuals assisting with the analysis of de-identified data, for example, would not be investigators under this definition.

Investigator Name

Rosenberg, David E

Role

Principal Investigator

System Role

Admin

Permissions

Full Access

USU Personnel Email

david.rosenberg@usu.edu

Department

Civil & Environmental Engineering

CITI Training Expiration Date

October 2, 2027

You only need to add personnel who will serve as an investigator on this project.

An **investigator** is an individual who interacts with or intervenes with living people for research purposes; obtains, studies, interprets, or analyzes identifiable private information; obtains informed consent; or interacts with the IRB relating to this study. Individuals assisting with the analysis of de-identified data, for example, would not be investigators under this definition.

Investigator Name

Osorio Diaz, Katherine Jeannette

Role

USU Student Researcher

System Role

Admin

Permissions

Full Access

USU Personnel Email

katherine.osorio@usu.edu

Department

Civil & Environmental Engineering

CITI Training Expiration Date

8/5/2027

You only need to add personnel who will serve as an investigator on this project.

An **investigator** is an individual who interacts with or intervenes with living people for research purposes; obtains, studies, interprets, or analyzes identifiable private information; obtains informed consent; or interacts with the IRB relating to this study. Individuals assisting with the analysis of de-identified data, for example, would not be investigators under this definition.

Investigator Name

Fager, Brittany

Role

USU Student Researcher

System Role

Admin

Permissions

Full Access

USU Personnel Email

brittany.fager@usu.edu

Department

Civil & Environmental Engineering

CITI Training Expiration Date

9/3/2028

You only need to add personnel who will serve as an investigator on this project.

An **investigator** is an individual who interacts with or intervenes with living people for research purposes; obtains, studies, interprets, or analyzes identifiable private information; obtains informed consent; or interacts with the IRB relating to this study. Individuals assisting with the analysis of de-identified data, for example, would not be investigators under this definition.

Investigator Name

Myers, Anabelle

Role

USU Student Researcher

System Role

Admin

Permissions

Full Access

USU Personnel Email
a02369941@usu.edu

Department

Civil & Environmental Engineering

CITI Training Expiration Date
2/5/2028

Does this project involve a Non-USU Investigator?

No

Categories of Exemption

Add all applicable exemption categories one at a time, by selecting "Add Line" or "Add Info" below.

Please select an exemption category. You may select another after this one has been saved.

EXEMPTION #2a: Surveys, interviews, educational tests involving adults

Project Overview

Provide an overview (or a brief abstract) for this study. The overview/abstract must include at least the following information:

1. The purpose of the study
2. The research questions, hypotheses, or themes this study will explore.
3. The "gap" or "problem" relevant to the discipline that this study will address.
4. How the study design will generate the information needed to address the research questions, hypotheses, or themes identified above.

The purpose of this study is to develop, use, and improve an immersive online collaborative model of a water bank for Cache valley, UT. We also intend collaborative model sessions to improve understanding of *why* collaborating basin partners choose to consume water and trade water with a water bank. We are also interested to learn *how* hydrologic, species interaction, social, economic, inclusion, uncertainty, and other elicited assumptions, factors, constraints, and opportunities influence collaborators choices to consume and bank water within the collaborative model environments. We also seek to learn *how much* model complexity and resolution collaborators request to achieve the study purposes.

Traditional water resources simulation and optimization modeling approaches answer the questions *what if* and *what is best*. Traditional simulation and optimization models also follow a project design where researchers build a model then translate the model and findings to stakeholders at the project end when there are no longer resources to follow up on stakeholder feedback. The build-translate project design represents a one-directional flow of information from researchers to stakeholders. We instead engage collaborators early in the modeling process. They take-away insights from collaboration in a model session. We use their feedback during a model session to improve the model assumptions, structure, and resolution. We anticipate feedback gained during collaborative model sessions will also yield a model that is more flexible and adaptive to user needs and challenges. Early engagement will also increase buy-in.

Upload a proposal here. Proposals are required for all Expedited and Convened IRB protocols, and for some Exempt protocols. If this study:

- is any review type other than exempt;
- has been proposed to an external funding source;
- is a dissertation/thesis project; or
- has been funded internally

You must provide the proposal that you submitted via that process, rather than a modified one for IRB purposes. The IRB is required to ensure that this protocol matches that project.

[Proposal-HAkbar_2024Nov_V2.pdf](#)

Upload one of the following:

- **Thesis Approval Form:** a PDF of your pending Thesis/Project Approval Form from ServiceNow. This form is likely pending, awaiting IRB approval, but if the committee has approved it, it can be provided here while pending.
- **Application for Candidacy:** a PDF of the Application for Candidacy from ServiceNow. This form is likely pending, awaiting IRB approval, but if the committee has approved it, it can be provided here while pending.
- **Signed Cover Sheet:** If your department utilizes signed cover sheets, provide one that has been signed by the student's committee.
- **Committee Member Communication:** A single PDF document containing communications from each committee member indicating that this project is likely to be approved by the student's committee, and is unlikely to require substantial changes upon proposal.

[Akbar-ProposalApproval.pdf](#)

Will any portion of this project be completed outside of the United States or in another jurisdiction that is sovereign, such as a territory or sovereign indigenous lands?

No

Please describe the setting(s) where this research project will occur.

Model sessions will take place during a video conference session (e.g., Zoom) or in-person in a university conference room. During a session, collaborators will enter choices to consume and bank water into a single, shared online numeric model of a water bank (e.g., web spreadsheets hosted in Box). We will prompt discussion of *why* collaborators made choices and *how* they will implement those choices.

Which response best describes the use of existing data (or biospecimens) about or from living people in this study?

Some existing and some new data/specimens will be used in this study

Study Participants

A participant is a living individual about whom an investigator conducting research obtains:

- 1) data or specimens through intervention or interaction with the individual; or**
- 2) identifiable private information or specimens.**

All participants must be described in this section.

Below, add a row for each distinct population that will be involved in your study. Select "+Add Line" (or "+Add Info") to add an individual row, and complete the information requested for each distinct population. You can add as many distinct populations as will be involved in your study.

Describe the general characteristics of these study participants.

Include a descriptive label, age range, health status, sex, gender, and any other characteristics that are relevant to your study. For studies involving only existing data or specimens, describe the populations that were involved in the original collection effort.

Experts and academics. Age: Over 21

Inclusion Criteria

Collaborators whose research is in water management, water markets, and/or water banks

Exclusion Criteria

N/A

How many participants from this group do you require in order to meet your study aims?

5-10

Will you be collecting information from or about people prior to obtaining informed consent in order to determine or verify eligibility?

No

Provide screening instrument(s) for this targeted population here. If you have more than one instrument, please combine them into one single document for this review.

[No Attachment](#)

Will all prospective participants have the potential to be recruited into and enter the study, regardless of sex, gender, sexual orientation, religion, race, color, adult age, genetic information, ability, nationality, or veteran status?

Yes

Identify the basis on which you must differentiate prospective participants, and provide the scientific justification for that differentiation.

Describe the general characteristics of these study participants.

Include a descriptive label, age range, health status, sex, gender, and any other characteristics that are relevant to your study. For studies involving only existing data or specimens, describe the populations that were involved in the original collection effort.

Practitioners. Age: Over 21

Inclusion Criteria

Collaborators who manage water systems or develop water policy

Exclusion Criteria

N/A

How many participants from this group do you require in order to meet your study aims?

5-10

Will you be collecting information from or about people prior to obtaining informed consent in order to determine or verify eligibility?

No

Provide screening instrument(s) for this targeted population here. If you have more than one instrument, please combine them into one single document for this review.

[No Attachment](#)

Will all prospective participants have the potential to be recruited into and enter the study, regardless of sex, gender, sexual orientation, religion, race, color, adult age, genetic information, ability, nationality, or veteran status?

Yes

Identify the basis on which you must differentiate prospective participants, and provide the scientific justification for that differentiation.

Describe the general characteristics of these study participants.

Include a descriptive label, age range, health status, sex, gender, and any other characteristics that are relevant to your study. For studies involving only existing data or specimens, describe the populations that were involved in the original collection effort.

Water Users. Age: Over 21

Inclusion Criteria

People who withdraw, distribute, or consume water on day-to-day basis.

Exclusion Criteria

N/A

How many participants from this group do you require in order to meet your study aims?

5-10 in each group

Will you be collecting information from or about people prior to obtaining informed consent in order to determine or verify eligibility?

No

Provide screening instrument(s) for this targeted population here. If you have more than one instrument, please combine them into one single document for this review.

[No Attachment](#)

Will all prospective participants have the potential to be recruited into and enter the study, regardless of sex, gender, sexual orientation, religion, race, color, adult age, genetic information, ability, nationality, or veteran status?

Yes

Identify the basis on which you must differentiate prospective participants, and provide the scientific justification for that differentiation.

Describe the study team's existing relationship to the participants who are targeted for recruitment into this study.

The study team (Akbar and Rosenberg) know some potential collaborators from past professional activities that span decades.

How many participants do you anticipate will begin this study?

30

How many participants do you anticipate will complete this study?

30

Will the research team offer compensation to any of the participants?

No

For each population group identified above, specify how the potential participants will learn about your study.

Participant Group Description

Experts and academics

Provide a detailed description of any recruitment processes that involve interactions with the prospective participants identified here.

The researchers will contact potential collaborators via email. See email template provided.

Participant Group Description

Practitioners.

Provide a detailed description of any recruitment processes that involve interactions with the prospective participants identified here.

The researchers will contact potential collaborators via email. See email template provided.

Participant Group Description

Water Users.

Provide a detailed description of any recruitment processes that involve interactions with the prospective participants identified here.

The researchers will contact potential collaborators via email. See email template provided.

Please select all of the following groups that are likely to be involved with your study.

No Vulnerable Populations

Provide all relevant recruitment material for this study below.

These must be developed in line with the IRB's [Recruitment Guidelines](#). You may upload as many as you need. Click "Add Info" or "Add Line" to access templates for many different kinds of recruitment materials.

Upload your recruitment material here. The IRB has several recruitment material templates or information items available to assist you. Click on the option you'd like to access, below:

- [SONA Registration Information](#) (Required if using SONA)
- [Amazon Mechanical Turk HIT Description](#) (Required if using MTurk)
- [FERPA Authorization](#) to Screen Education Records for Eligibility
- [HIPAA Authorization](#) to View Protected Health Information for Eligibility
- [Email Recruitment Best Practices](#)

If you have no recruitment material, please upload a placeholder document indicating that you have no materials. In most cases, you would at least use a Letter of Information or script, so this should be very uncommon (i.e. only in protocols using existing data).

[StudyInformation-PostitBoard-April2025.docx](#)

Recruitment Material Description

Online poster board that provides potential collaborations with information about the study so that they can make an informed decision for whether to participate in the study.

Upload your recruitment material here. The IRB has several recruitment material templates or information items available to assist you. Click on the option you'd like to access, below:

- [SONA Registration Information](#) (Required if using SONA)
- [Amazon Mechanical Turk HIT Description](#) (Required if using MTurk)
- [FERPA Authorization](#) to Screen Education Records for Eligibility
- [HIPAA Authorization](#) to View Protected Health Information for Eligibility
- [Email Recruitment Best Practices](#)

If you have no recruitment material, please upload a placeholder document indicating that you have no materials. In most cases, you would at least use a Letter of Information or script, so this should be very uncommon (i.e. only in protocols using existing data).

[Recruitment Email-April2025.docx](#)

Recruitment Material Description

Recruitment email.

Upload your recruitment material here. The IRB has several recruitment material templates or information items available to assist you. Click on the option you'd like to access, below:

- [SONA Registration Information](#) (Required if using SONA)
- [Amazon Mechanical Turk HIT Description](#) (Required if using MTurk)
- [FERPA Authorization](#) to Screen Education Records for Eligibility
- [HIPAA Authorization](#) to View Protected Health Information for Eligibility
- [Email Recruitment Best Practices](#)

If you have no recruitment material, please upload a placeholder document indicating that you have no materials. In most cases, you would at least use a Letter of Information or script, so this should be very uncommon (i.e. only in protocols using existing data).

[ImmersiveModelingBearRiver-March2026.docx](#)

Recruitment Material Description

Compiled document of all prior approved and 2 new requested invitation methods. Includes more detail on the why and how of our study and an example summary of results to date to include in technical presentations/posters/etc.

Study Procedures

Provide a complete, step-by-step description of the study procedures.

1. Contact potential collaborators via email. Provide link to study information and informed consent (online poster board).
2. Ask collaborators to sign up for a model session.
3. The researchers will group consenting collaborators and set up online sessions.
4. Collaborators collaborate in a 2-hour model session. They enter choices to consume and bank water in response to modeled available water, others choices, and real-time discussion of rationales for choices.
5. We ask collaborators why they made choices, which features of the water system are correctly represented in the model, how can we improve the model.
6. Researchers take notes on the discussion, including rationales for choices and feedback to improve the model.

The Excel model file is attached. As part of our research method, we will update and improve the model in response to feedback from collaborators and use improved models in future model sessions.

Please specify your anticipated end dates for the items below. These dates will be used to set the expiration date for your informed consent documents (if applicable) and your protocol, so be

certain to keep this information up-to-date as your work progresses.

By what date do you anticipate that your protocol will no longer actively involve a human subject?

June 30, 2027

By what date do you anticipate all study-related procedures (analyses, follow ups, etc.) will be complete?

December 31, 2028

What measures will be taken to ensure participant privacy during the study?

Consider procedures from recruitment through the end of the participant's involvement in the study.

Privacy means the extent to which participants have control over what they share, with whom they share it, and in what kinds of spaces. Do not address confidentiality of the data - that will be addressed later.

Like other professional activities, all collaborators who participate in the same model session will be able to view entries into the web spreadsheet and hear the real-time discussion of choices and feedback.

Will there be any email communications with participants during the course of this study? This includes email communications regarding recruitment, informed consent, data collection, compensation, scheduling, etc.

Yes

Please make the appropriate selections below to indicate your understanding of appropriate email communication practices.

Check this box to indicate that you understand that all university business must be conducted with a usu.edu email address.

Check this box to indicate that you will use "bcc" if you must communicate with several participants at once, unless your informed consent document specifically discloses that participants will be communicated with in groups.

Earlier, you indicated that existing data would be utilized in this study. Please identify below:

1. The source of the existing data or specimens (including the specific custodian of the data/specimens)
2. A description of the existing data or specimens that will be used
3. How the existing data was or will be acquired by the research team

Completed model files and notes from model sessions taken by Dr. Rosenberg and Hadia Akbar during sessions conducted in Fall 2023 before we identified the need to complete a human subjects protocol. These electronic files are already in possession by Dr. Rosenberg and Hadia Akbar. The files are stored in USU's Box system and handled with the same methods as new-to-collect data listed in this protocol.

Will this study involve the use of any machines, devices, or substances supplied by the research team that the participants directly interact with?

Yes

Please select "Add Info" to provide details about the equipment, substances, or devices that participants will interact with as a part of your research project.

Equipment, machine, device, or substance name:

Online web spreadsheets.

Equipment, machine, device, or substance manufacturer:

N/A

Describe the equipment, machine, substance, or device:

N/A

Describe how participants will interface with this equipment, substance, or device:

Through their internet browser.

If applicable, provide a link to the product brochure, specification sheet, manufacturer description, sales information, or other printed material from the manufacturer (or provide an upload in the next field).

If no link is available, please provide a product brochure, specification sheet, manufacturer description, sales information, or other printed material from the manufacturer.

[No Attachment](#)

Equipment, machine, device, or substance name:

Online webspreadsheet for Bear River Model

Equipment, machine, device, or substance manufacturer:

NA

Describe the equipment, machine, substance, or device:

NA

Describe how participants will interface with this equipment, substance, or device:

Through their web browser

If applicable, provide a link to the product brochure, specification sheet, manufacturer description, sales information, or other printed material from the manufacturer (or provide an upload in the next field).

If no link is available, please provide a product brochure, specification sheet, manufacturer description, sales information, or other printed material from the manufacturer.

[BearRiverWaterBank_Feb2026.xlsx](#)

Will this study involve the collection, use, or analysis of any of the following?

- Images of participants or others
- Audio recordings of participants or others
- Video recordings of participants or others

No

Please upload all measures, instruments, data collection sheets, and relevant intervention materials here. If a participant will be asked to interact with something designed or used by your research team, it must be provided here.

Choose a document to upload here

[CacheValleyWaterBank_2024Feb.xlsx](#)

Provide a brief description of this document

Current version of spreadsheet model file we will use during a collaborative model session.

Choose a document to upload here[List of Questions.docx](#)**Provide a brief description of this document**

List of questions to ask collaborators during a collaborative model session.

Choose a document to upload here[BearRiverWaterBank_Feb2026.xlsx](#)**Provide a brief description of this document**

Current version of spreadsheet model file we will use during a collaborative model session (Bear River basin scale)

Confidentiality

Check all of the following applicable data/specimen sources that will be accessed, obtained, created, or used in this study. This includes information you will use to recruit or screen.

None of the above

Does this project involve identifiable information or specimens?

Identifiable means that the identity of the living person "is or may readily be ascertained." Some examples include:

- Email addresses
- Telephone numbers
- Recordings (audio or video) or images
- Codes linking back to an identity that the research team *could* access (regardless of whether the research team *will* access it)
- Transcriptions that have not been scrubbed of identifying information

Yes

Describe which identifying information will be present for which data/specimens.

Be sure to include information you will have for recruitment, compensation, screening, informed consent processes (not the document), etc.

- Participant Email addresses: for recruitment and to share letter of information.
- Participant Names: During a model session, participants will enter their name in the water user's section of the model. The meeting notes will contain name information as well.
- After a model session, we will update the names in the Water Users section of the model according to the wishes of the collaborator (e.g., anonymous, by job title, organization, etc.).
- We will change the permissions on the model file to only be viewable and editable by the research team.

Explain where the data, information, or biospecimens will be housed *during the data collection process*.

Outline applicable confidentiality safeguards associated with that initial data storage mechanism(s).

- The recruitment material, model file, and meeting notes containing feedback will be kept on USU Box drive.
- We cannot guarantee confidentiality for participants names, rationales for choices, and feedback as other collaborators in the same session will view choices and hear discussion

After the data, information, or biospecimens have been obtained by the research team, what will happen to prepare for and begin use/analysis? *Be sure to include how data will be moved from a temporary collection site to longer-term storage, what will be removed or coded, what will be deleted or maintained, and what storage repositories will be utilized.*

- Collect feedback to improve the model.
- Use feedback to improve the model.
- Remove names and other identifying information from model files and notes from sessions. Set permissions on the model file to only be accessible by the research team.
- Post cleaned (anonymous) versions of model files and notes on a permanent repository such as Hydroshare.

Describe the extent to which confidentiality will be maintained during the study, and after it has ended. What steps will you take, if any, to protect participants' identities at each of those time points?

1. During collaborative model sessions, collaborators will have access to view and edit the current version of model. This version will include collaborator's names and their choices.
2. At the end of a model session, collaborators will choose how they wish to be identified (anonymous, by organization, by name, etc.). The research team will respect those choices. We will set permissions on the model file so only the research team has access to the model file (view and edit).
3. Like other professional activities – collaborators participating in the same model session will be able to see other participants choices and listen to their rationales.

Legal structures exist throughout the U.S. and beyond that require the reporting of certain behavior or conditions. Items that require reporting include, but are not limited to, the following:

- Child abuse and neglect
- Abuse, neglect, and exploitation of a vulnerable adult
- Sexual assault
- Specific imminent threats to self or others
- Certain medical conditions under public health surveillance oversight

During the conduct of this research, is any member of this research team in a position to learn about one or more circumstances which requires mandatory reporting or other breaks from typical confidentiality protections (such as to disclose a health condition to a third party)?

No

Use the table below to describe the information, data, or specimens that you will keep beyond the completion of this study, and those that you will destroy at the completion of this study. Click "Add Line" or "Add Info" below.

Once the project has reached [completion](#), what data/information/biospecimens, if any, will be maintained by the research team or others? Where will the information be maintained? If, for example, you plan to keep de-identified data or participant contact information beyond the term of this study, be sure to list it here.

De-identified versions of start and completed model files for each model session will be posted on a permanent repository such as Hydroshare. De-identified versions of notes from each model session will also be posted on a permanent repository such as Hydroshare.

Once the project has reached completion, what data, if any, will be destroyed?

Versions of completed model files with collaborators names. Versions of notes from model sessions with collaborators names.

Informed Consent - Exemptions

Informed Consent refers to two things: first, the *process* of obtaining and maintaining the fully informed consent of research participants. The process is ongoing throughout the lifetime of the study, and can take place in a wide variety of ways. Second, Informed Consent refers to the *documentation* used to explain and obtain agreement for study participation.

This section will ask the research team to consider both the *process* and *documentation* relating to informed consent for this study. Many exemptions require you to obtain the prospective agreement of the participants in the study, even though the project otherwise qualifies for exemption, so the questions here are not as specific as would be under the Expedited categories of review.

Describe both the process and the setting for obtaining informed consent from potential participants. If the process and setting differs for different groups, be sure to separately address each group.

Process

1. Research team will invite collaborators to participate in the study via email.
2. The email will link to an online poster board that provides information about the study. The poster board will provide all information a potential collaborator needs to consent to participate in the study – study title, goals, what is requested of collaborators, data to collect, how stored, risks, possible benefits, how to ask questions. At the bottom of the board, there is the text:

By attending a model session, you indicate that you understand the risks and benefits of participation. You also acknowledge that you know what you will be asked to do. You had the opportunity to ask questions. And you are clear on how to stop participation if you choose to.

3. Researchers will provide potential collaborators a list of dates/times for potential model sessions.
4. Collaborators will select a model session(s) to participate in.

Documentation:

5. During a model session, collaborators will enter their name in the online model file (web spreadsheets).

Upload your informed consent documentation here. You may upload as many as needed by clicking "Add Line" after each successful upload. The IRB has developed several

templates for informed consent documentation that you are encouraged to use. [Access all of them here](#). The USU IRB has a template specifically for use in exempt studies, which you may [access for download directly at this link](#).

Upload your informed consent documentation here.

Your upload must be in Word. If it is not, your protocol will be returned to you during the pre-review process.

[StudyInformation-PostitBoard-April2025.docx](#)**Upload your informed consent documentation here.**

Your upload must be in Word. If it is not, your protocol will be returned to you during the pre-review process.

[Recruitment Email-April2025.docx](#)**Upload your informed consent documentation here.**

Your upload must be in Word. If it is not, your protocol will be returned to you during the pre-review process.

[Email-PriorParticipants.docx](#)**Upload your informed consent documentation here.**

Your upload must be in Word. If it is not, your protocol will be returned to you during the pre-review process.

[ImmersiveModelingBearRiver-Feb2026.docx](#)

Conflicts & Assurances

Conflict of Interest

Please review the definition of a [conflict of interest](#).

I have reviewed and I understand the definition of a Conflict of Interest

I, the Principal Investigator, affirm that I do not have any conflicts of interest regarding this project.

I, the Principal Investigator, affirm that I have reviewed the definition of a conflict of interest with the research team. No one on this research team has a conflict of interest.

Does this protocol involve the use of any intellectual property owned or held by Utah State University?

No

Principal Investigator's Assurances

As Principal Investigator, I have ultimate responsibility for the performance of this study, the protection of the rights and welfare of the human subjects, and strict adherence by all co-investigators and research personnel to all Institutional Review Board (IRB) requirements, federal regulations, and state statutes for human subjects research. Every aspect of the research will have appropriate and consistent oversight by the Principal Investigator. I hereby assure the following:

- I will not enroll or recruit participants in research until IRB review and approval of this study has been completed;
- The information provided in this application is accurate to the best of my knowledge;
- Adequate resources are available to carry out this research and ensure participants' safety and welfare;
- All named individuals on this project have read and understand the procedures outlined in the application;
- All experiments and procedures involving human subjects will be performed under my supervision or that of another qualified professional listed on this protocol.

DER

I and all co-investigators and research personnel agree to comply with all applicable requirements for the protection of human subjects in research, including but not limited to the following:

- Making no changes to the approved protocol without first having submitted those changes for review and approval by the IRB (except as necessary to protect participants in an emergency);
- Promptly providing the IRB with any information requested related to the project;
- Promptly reporting the completion of a study and requesting [closure](#), including ensuring that I maintain sufficient oversight of student projects to appropriately report on the project for closure;
- Disclosing all [reportable events](#) to the IRB as soon as possible;
- Promptly and completely complying with an IRB decision to suspend or withdraw its approval for the project;
- Permitting and facilitating any Utah State University Human Research Protection Program site visits, audits, investigations, and inquiries, including access to all research-related documents;
- Obtaining renewal prior to the expiration of this protocol's approval, including ensuring that I maintain sufficient oversight of student projects to seek renewal in the event the student(s) become unavailable. I understand that timely renewal is my responsibility, and if I fail to apply for renewal, the study will expire and all study activity must cease until current IRB approval is obtained.

DER

I assume responsibility for ensuring the competence, integrity, and ethical conduct of all members of the research team. This research project is being proposed on behalf of at least one USU employee or student, to be completed in their capacity as a USU employee or student. I certify that all researchers are fully competent to accomplish the procedures and tasks assigned to them during the completion of this research project. All researchers will maintain current CITI and any other applicable training for the entirety of their involvement with this project.

DER

Please explain how often and in what ways the PI will monitor this student project.

Among other responsibilities articulated in [Standard Operating Procedure 302](#), Principal Investigators are expected to: delegate only those study procedures which are appropriate for delegation and ensure that delegated tasks are completed by an individual with appropriate training, skills, and experience; have and maintain access to study data throughout the course of the study and after its completion; actively monitor the student's progress and any potential harms or adverse events to participants; ensure timely processing of participant compensation; and ensure that, at all times, the protocol is being carried out with fidelity to the requirements in all relevant jurisdictions.

I will monitor continually as a co-collaborator on Hadia's project to ensure the protocol is followed.

Is there any other information that you would like to provide to the IRB as it works through the review of your project? For example:

- School district approval letter (required for all work in K-12 schools in Utah);
- Tribal IRB, other IRB/REC, or international ethics committee approval;
- Letters of support;
- Evidence of scientific, scholarly, or peer review that has already occurred; or
- Other documents, links, or informational items that would support the IRB's review of this project.

Yes

Other Approvals or Documentation

File Upload

[RosenbergcitiCompletionCertificate_1182028-October2-2024.pdf](#)

Upload description

CITI Certification - Rosenberg - October 2, 2024

File Upload

[RosenbergcitiCompletionReport_1182028_-October2-2024.pdf](#)

Upload description

CITI Completion reprot - Rosenberg October 2, 2024

The PI may submit this protocol now, using the menu on the right side of the screen. The PI must review all information contained in this form (initial and resubmissions) prior to submitting to the IRB.

Administrative Details Form

Administrative Protocol Details

Is there a pending Thesis Approval Form or Application for Candidacy that needs to be signed when this protocol's review is complete?

No

Please select the type of review being initiated. Amendment

If the type is already selected, do not unselect it, just proceed to that table and populate your entry.

Exempt

Exemptions

Below, log the date on which the action occurred for this exempt review. Please leave actions which have not occurred or which have not occurred yet blank.

Date on which this exemption was submitted.

March 4, 2024

Date on which the pre-review was completed for this protocol and a review type was assigned.

No date entered

Date on which a determination or resolution was made regarding this exemption

March 4, 2024

What determination completed this review cycle?

Return to Researcher - did not pass pre-review

What is the next step for this submission?

Select the appropriate categories of exemption for this protocol

Date on which this exemption was requested closed or canceled by the research team.

No date entered

Date on which this exemption was closed or cancelled

No date entered

Was this protocol closed and completed, or was the project cancelled?

Admin Comments regarding the Exempt review:

Below, log the date on which the action occurred for this exempt review. Please leave actions which have not occurred or which have not occurred yet blank.

Date on which this exemption was submitted.

March 6, 2024

Date on which the pre-review was completed for this protocol and a review type was assigned.

March 7, 2024

Date on which a determination or resolution was made regarding this exemption

March 11, 2024

What determination completed this review cycle?

Exempt

What is the next step for this submission?

Select the appropriate categories of exemption for this protocol

Exempt 2: Educational tests, survey procedures, interview procedures, or observation of public behavior where the disclosure of responses outside the research would not place the participants at risk

Date on which this exemption was requested closed or canceled by the research team.

No date entered

Date on which this exemption was closed or cancelled

No date entered

Was this protocol closed and completed, or was the project cancelled?

Admin Comments regarding the Exempt review:

Date on which this amendment was submitted

August 6, 2024

Date on which a determination was rendered regarding this amendment

August 6, 2024

What determination was rendered for this amendment?

Approved - Expedited Minor Modification

Admin notes regarding this amendment review

Date on which this amendment was submitted

October 2, 2024

Date on which a determination was rendered regarding this amendment

October 2, 2024

What determination was rendered for this amendment?

Approved - Expedited Minor Modification

Admin notes regarding this amendment review

Date on which this amendment was submitted

December 11, 2024

Date on which a determination was rendered regarding this amendment

December 11, 2024

What determination was rendered for this amendment?

Approved - Expedited Minor Modification

Admin notes regarding this amendment review

Date on which this amendment was submitted

April 8, 2025

Date on which a determination was rendered regarding this amendment

April 8, 2025

What determination was rendered for this amendment?

Approved - Expedited Minor Modification

Admin notes regarding this amendment review